

# Research on the Application of Big Language Models in Cybersecurity Incident Response—A Log Analysis Approach Based on GPT Models

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**Abstract**—In the face of increasingly complex and high-frequency cyberattacks, cybersecurity incident response (IR) is facing the challenges of both efficiency and accuracy. The traditional process relies on manually analysing a large number of logs, which is both time-consuming and error-prone. In recent years, Large Language Models (LLMs) with powerful natural language processing capabilities have brought new opportunities for automated security analysis. In this paper, we propose a cue engineering-driven log analysis method based on GPT-3.5 Turbo, focusing on two key tasks: attack stage classification and event summary generation. Experimental results based on the CIC-IDS2017 dataset show that the method achieves an overall accuracy of about 40% in attack phase identification, and is particularly accurate in the Reconnaissance phase; and the ROUGE-L metric improves by about 12% compared to the keyword matching method in the log summary task. In addition, this paper systematically analyses key issues such as model illusion, uncertain output and data privacy, and makes suggestions for improvement. This study verifies the application prospect of big language models in cybersecurity incident response, and provides theoretical and practical support for building intelligent and secure response processes.

**Keywords**—cyber security, incident response, large language modelling, GPT, log analysis, MITRE ATT&CK

## I. INTRODUCTION

In the current highly interconnected digital environment, cyber-attacks are becoming more frequent, presenting the characteristics of larger scale, more covert methods, and more complex attack chains as the core link in responding to intrusion, containing loss and restoring service, cyber security incident response (IR) is facing unprecedented challenges. Efficiently and accurately analyzing security logs and quickly identifying attack stages and paths is the key to improving incident response efficiency.

The current incident response process still relies on manual analysis and rule-based detection systems (e.g., security information and event management system SIEM, signature-based intrusion detection system IDS), with the following significant limitations:

**Low efficiency:** The analysis process is highly dependent on manual labor, which is time-consuming and prone to missing judgments when dealing with a large number of logs;

**Poor response:** limited ability to identify new and uncharacterized attacks;

**Difficult to maintain:** System rules must be updated

frequently, and high maintenance costs and lagging response.

With the development of AI technology, more and more methods based on machine learning (ML) and deep learning (DL) have been applied to the field of cyber security to enhance the automation of threat detection and incident response. For example, studies have used supervised learning classifiers to identify anomalous traffic or sequence models such as LSTM to analyze log patterns. However, these approaches usually rely on large amounts of high-quality labeled data, and their generalization ability is insufficient to cope with the continuous evolution of complex attacks.

In recent years, Large Language Models (LLMs), such as GPT-3.5 and GPT-4, have demonstrated excellent natural language understanding and generation capabilities and are able to deal with a mixture of structured and unstructured log data. Compared with traditional methods, LLMs can perform tasks such as log analysis, attack stage classification, and event summarization through Prompt Engineering without specialized training, which has the following advantages:

Understanding natural language logs to quickly infer attack intent;

Migration capabilities to recognize unseen attack patterns;

Automatic generation of logically coherent and readable analysis summaries.

Although related research has initially explored the application of LLMs in the direction of vulnerability scanning, malicious code analysis, and security policy generation [2–4], there is still a lack of systematic research and unified evaluation framework for log-level attack stage identification and summary generation in security incident response scenarios. This research gap limits the effective deployment and evaluation of large-scale language models in actual security operations.

To address these issues, this paper proposes a log analysis method based on GPT-3.5 Turbo, focusing on two key tasks:

1) Attack stage classification based on the MITRE ATT&CK framework [1];

2) Natural language summary generation of security events.

The methodology adopts a structured cue engineering process and conducts experimental evaluations based on the CIC-IDS2017 public dataset [8]. It systematically analyses the model's performance in terms of classification accuracy, summary quality, and comparison of traditional methods to

reveal its strengths and limitations.

The main contributions of this paper are as follows:

Designing cue templates applicable to log analysis scenarios, enabling attack stage identification and event summarization;

Constructing a reproducible experiment and evaluation process to validate the effectiveness of the method on public datasets;

Evaluating model performance in phases and analyzing various types of misclassifications and applicable boundaries in depth;

Exploring practical application issues such as model illusion, uncertainty output, and data privacy;

Looking forward to the future development direction of combining fine-tuning optimization, local deployment, and rule model fusion.

The overall flow of the proposed method is shown in Fig. 1, covering the whole process from raw log preprocessing, structured cue design, and model inference invocation to classification and summary generation in the attack phase.

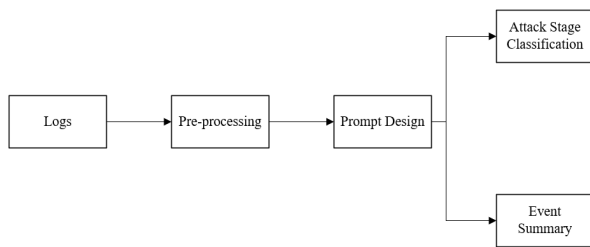


Fig. 1. Architecture of GPT-based Incident Response Log Analysis System

## II. METHODOLOGY

### A. Overall Process Overview

To achieve network security log analysis based on large-scale language models, this paper constructs a complete set of analysis processes, which contains four main steps: log data preprocessing, prompt design (Prompt Engineering), model inference invocation, and output post-processing [4]. The overall framework is shown in Fig. 2.

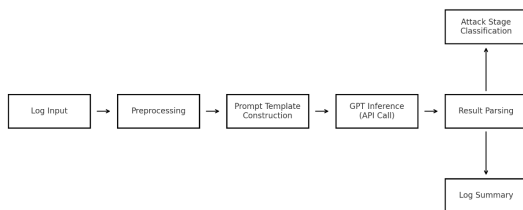


Fig. 2. Log Analysis Process Based on GPT Model

### B. Log data pre-processing

In order to improve the large-scale language model's understanding of network logs, it is necessary to preprocess the raw data, which mainly includes the following steps:

Field Extraction and Cleaning: Retain core fields such as timestamps, IP addresses, protocols, event descriptions, etc., and eliminate redundant and noisy information;

Terminology standardization: Expand common network terms and abbreviations (e.g., DNS, SYN Flood) into standard expressions to reduce model misinterpretation;

Natural language formatting: rewrite structured logs into natural language sentences to make them more consistent with model language conventions;

Length control: Ensure that each log text is within the GPT-3.5 Turbo maximum input length (~4,096 tokens).

### C. Prompt Design for Attack Phase Classification

To effectively classify attack phases based on the MITRE ATT&CK framework, this paper introduces a structured prompt template that assists the model in making accurate decisions within a predefined list of phases. The prompt template consists of three main components:

Task description: guides the model through a classification task, e.g., 'Please read the following log snippet and determine the stage of attack it is in.'

Stage List: Refer to the attack stages provided by MITRE ATT&CK, including Reconnaissance, Initial Access, Execution, Persistence, Privilege Escalation, Exfiltration, Command and Control;

Log content: insertion of pre-processed log text that constitutes the model input.

Example prompt format:

Log Fragment: [Log Text]. Please select the category that best matches the following phases: Reconnaissance, Initial Access, Execution, Persistence, Privilege Escalation, Exfiltration, Command, and Control.

Model setup: The OpenAI GPT-3.5 Turbo model is invoked with a temperature set to 0.2 to improve output stability [3]. Each log is reasoned individually, and the output includes the prediction stage and (optionally) a reasoning statement.

### D. Prompt Design for Log Summarisation Generation

To facilitate the generation of natural language summaries for security events, this paper presents a collection of prompt templates designed to assist the GPT model in creating concise summaries with coherent content and key points.

Task description: e.g., 'Please summarise the event's main attack behaviors with the attacker's possible objectives based on the following log content.'

Log input: Use formatted natural language log text.

Example prompt format:

Log snippet: [log text]. Please summarise the incident's key behaviors and the attacker's possible aims, requiring concise language and highlighting key points.

Inference parameter setting: Use the GPT-3.5 Turbo model and control the summary length between 3080 words. The hints explicitly prohibit the model from making unsupported inferences to enhance the credibility of the output.

### E. Model Output Post-processing

Given the randomness and uncertainty of the large language model in generating the content, in order to ensure the fairness of the evaluation and the reliability of the results,

this paper standardizes the model output as follows:

**Attack stage output standardization:** if the stage name returned by the model does not match the predefined list exactly, it is normalized by the string nearest-neighbor algorithm; those that cannot be matched are uniformly classified into the 'Unknown' category;

**Abstract consistency review:** Manual sampling is used to check whether the abstracts are consistent with the original log facts, and fictitious or logically incorrect information is eliminated;

**Low-confidence marking:** Output content containing expressions such as 'maybe...' and 'suspected to be...' are marked as low-confidence samples and recorded separately for subsequent analysis.

### III. EXPERIMENTS & EVALUATION

In order to evaluate the feasibility and effectiveness of the proposed large language model log analysis method in real network security scenarios, a series of experiments are designed and implemented in this paper. In this section, the experimental setup, evaluation metrics, and experimental results for the classification and summarisation tasks are presented in detail and analyzed through graphs and charts in comparison with the baseline approach.

#### A. Experimental Setup

This study conducts experiments using the CIC-IDS2017 public dataset. This dataset simulates normal network traffic as well as various types of attacks, including port scans, denial of service (DoS), brute force attacks, and botnets, within real enterprise network environments. It offers a wealth of log samples for research into security incident response.

To carry out the attack stage classification task, we manually label the attack stages to which selected log entries belong based on the attack scripts and event labels provided in the dataset concerning the seven stages in the MITRE ATT&CK framework [1] (Reconnaissance, Initial Access, Execution, Persistence, Privilege Escalation, Exfiltration, Command and Control), with an additional 'Unknown' category to cover cases that cannot be recognized or inferred by the model.

For the model, OpenAI GPT-3.5 Turbo was used for inference via API. Each log sample calls the model independently for attack stage prediction and event summary generation.

#### B. Assessment metrics and baseline methodology

##### Attack Stage Classification Tasks

The following standard classification metrics are used:

Accuracy Precision Recall F1-score

The confusion matrix also analyzes the model's classification performance and confusion trends at different stages.

##### Log summary task

Due to the lack of a standard answer, the summary task is evaluated in two ways:

**Automatic assessment metric:** ROUGE-L is used to measure the coverage concerning the reference summary;

**Subjective expert assessment:** 3 cybersecurity researchers

score the model output for relevance and completeness.

As a baseline for comparison, we introduce a rule-based approach based on keyword matching: by mapping feature words in logs (e.g., 'nmap,' 'SYN flood,' 'DNS query') in the logs to map to predefined attack phases or event summaries to mimic the rule engine logic in traditional SIEM/IDS systems.

#### C. Analysis of Attack Phase Classification Results

The GPT model achieves an overall accuracy of about 40% in the attack phase classification task. Among them, the most outstanding performance is in the Reconnaissance phase, which can effectively classify typical scans, host probes, and other logs correctly.

As shown in Fig. 3, the model's predicted distribution in the Reconnaissance phase is highly consistent with the proper labels. In contrast, in other phases (e.g., Initial Access, Execution), there is a significant deviation, which is often incorrectly classified as Unknown [9]. This suggests that the model tends to be conservative in its output in the absence of obvious semantic cues.

Figure 4 illustrates the confusion matrix for the classification of the attack phase. The reconnaissance phase has a high degree of aggregation on the diagonal and stable performance. In contrast, the Initial Access phase has a poor prediction effect and a high misclassification rate and is mainly confused with Unknown and Execution, which reflects the insufficiency of the GPT model's ability to identify the features of covert attack patterns. The GPT model achieves an overall accuracy of about 40% in the attack phase classification task. The most outstanding performance is in the Reconnaissance phase, which can effectively classify typical scans, host probes, and other logs correctly.

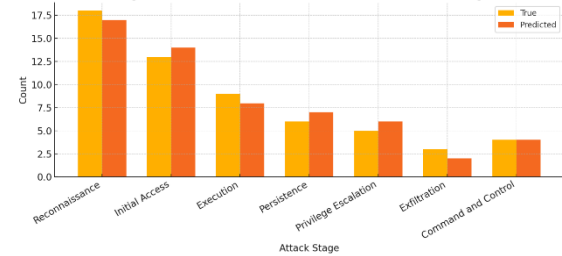


Fig. 3. Distribution of True vs GPT-Predicted Attack Stages

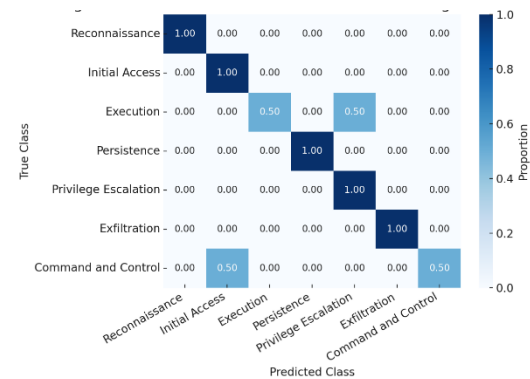


Fig. 4. Confusion Matrix of GPT Model Attack Stage Classification

#### D. Analysis of Log Summary Generation Results

In the log summarization task, the GPT model demonstrates strong linguistic organization and semantic summarization capabilities [10]. It can generate natural

language summaries with a coherent structure and precise logic in most scenarios. For example, for the DNS query failure log, the model outputs, 'The attacker attempted to resolve the domain name to gather intelligence, and the behavior may belong to pre-reconnaissance.' This demonstrates the ability to understand contextually and attack intent inference.

TABLE I. COMPARISON OF EXAMPLE LOG SUMMARY GENERATION (GPT VS. KEYWORD RULE)

| Log Entry                                                                                 | GPT Summary                                                              | Keyword-based Summary                               |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------|
| 192.168.1.10 sends multiple SYN requests to external IPs, triggering scan detection rule. | Detected mass SYN scan from 192.168.1.10, suspected network scanning.    | Multiple SYN requests, external IP, scan detection. |
| 172.16.0.5 frequently communicates with external DNS servers, possible DNS tunneling.     | Possible DNS tunneling by 172.16.0.5 due to frequent DNS communications. | Frequent DNS interactions, possible DNS tunneling.  |

In contrast, the rule-based approach, although deterministic, can only cover logs containing feature keywords and has limited ability to handle non-obvious attack patterns.

Table I compares the two examples. It can be seen that GPT outputs summaries with more contextual integrity and logical fluency. At the same time, the rule-based approach only extracts keywords and lacks the explanatory nature of attack behaviors.

On the ROUGE-L metric, GPT improves by about 12% compared to the rule-based method, indicating its advantage in summary coverage and information integration capability [10].

#### IV. DISCUSSION

The experimental results show that the large-scale language model based on GPT-3.5 Turbo has some potential for cybersecurity log analysis, especially regarding interpretability and text generation quality. However, several key limitations are also revealed, worth exploring in depth.

##### A. Inadequate classification accuracy

Although the model performs better in the reconnaissance phase, the recognition accuracy in the attack phases, such as Initial Access and Execution, is significantly low. The main reasons may include:

Lack of professional samples in the corpus: The pre-training data of the GPT model contains fewer semantics of the attack phases of structured security logs, which makes it difficult for the model to understand certain professional attack patterns [3~5] accurately;

Log features are not explicit: the log content corresponding to some stages (e.g., initial access) is more neutral and lacks obvious attack semantic features, which increases the difficulty of model judgment;

Conservative model strategy: When the hints are unclear or semantically ambiguous, GPT tends to output the 'Unknown' category to avoid misclassification.

In order to improve the classification performance, two strategies can be considered: one is to enhance the contextual information in the hints, such as introducing the definition of

attack stages with typical behaviors as an aid; the other is to use a language model that is fine-tuned explicitly on the security corpus in order to improve the comprehension ability in the professional domain.

##### B. Interpretability and Consistency Issues

Although GPT can generate explanatory text, these explanations may not necessarily be logically rigorous or consistent with the attack's principles. There are cases of 'seemingly reasonable but wrong.' Such pseudo-interpretation may mislead security analysts and reduce the system's credibility.

In addition, due to the generative model's randomness, the output results may differ even if the same log is analyzed multiple times [4]. This inconsistency is unacceptable in the security response process.

Possible improvement options include:

Freezing cue templates and model parameters to improve output stability;

Introducing a rule-assisted mechanism for post-validation of model output to ensure logical and semantic consistency;

Employing model integration strategies or confidence thresholding strategies to control output fluctuations in critical tasks.

##### C. Data Privacy and Response Latency Issues

In practical deployments, enterprise security logs often contain sensitive asset information, and uploading them to the cloud LLM model poses a risk of privacy leakage [6]. In addition, the network latency introduced by API requests is detrimental to real-time response requirements.

To this end, consider:

Deploying local large language models (e.g., LLaMA, Mistral) to ensure that data does not leave the intranet;

Implementing log desensitisation pre-processing (e.g. IP anonymisation);

Designing a hybrid local+cloud architecture leaves sensitive parts to be analyzed locally and complex tasks handled by the cloud model.

##### D. Inadequate summary assessment criteria

The log summary task relies mainly on manual subjective judgment and lacks uniform objective assessment criteria [9]. Although ROUGE-L is used, its relevance to the analysis results is still doubtful.

The following directions can be explored in the future:

Design and train a security event summary quality assessment model for automatic scoring;

Introducing a user research mechanism to measure the usefulness of summaries for analyst assistance;

Developing specialized assessment metrics that match the logs' structural characteristics to enhance the metrics' confidence.

#### V. CONCLUSION

This paper proposes a large-scale language model web log analysis method based on GPT-3.5 Turbo, constructs structured cueing templates and inference processes for two

key tasks in cybersecurity incident response: attack stage classification and event summary generation, and carries out a systematic empirical evaluation based on the CIC-IDS2017 dataset.

The experimental results show that the method has excellent recognition performance in the Reconnaissance phase, with strong contextual understanding and inference capability. Meanwhile, the generated natural language summaries outperform the traditional keyword matching method in terms of semantic integrity and readability, with an improvement of about 12% in ROUGE-L metrics, which verifies the actual potential of LLM in the task of security log understanding.

However, the model has a limited recognition effect in initial access and privilege elevation phases and suffers from limitations such as high output randomness, loose interpretation, and privacy sensitivity, which limits its direct deployment in real-world scenarios.

Based on the current research results and shortcomings, in-depth exploration can be carried out in the following directions in the future:

1) Model optimization and fine-tuning: Try to introduce security domain fine-tuning models (e.g., SecLLM) [4~5] or migrate to more powerful models, such as GPT-4/Claude, in order to improve the recognition of complex attack semantics;

2) Cue template adaptive design: using automatic cue search (AutoPrompt, Prompt Tuning) technology to dynamically optimize the cue structure to improve classification accuracy and stability;

3) Rule and LLM fusion analysis framework: build a collaborative analysis system between the rule engine and language model, combining the semantic understanding of LLM and the determinism of the rule method to improve the robustness of the overall system;

4) Local deployment and privacy control: exploring lightweight deployment schemes (e.g., quantized LLaMA), combining log desensitization technology and enterprise intranet calls to achieve private deployment of models;

5) Expanded dataset and multi-language adaptation: introduce richer attack scenarios and multi-language log samples to improve the model's generalization capability and internationalization adaptability;

6) Integration with SOC system: This research method

will be embedded into the actual security operation center (SOC) process to collect feedback and optimize model interaction and response efficiency.

In summary, although the current methodology still needs improvement, this study provides a feasible path and empirical evidence for applying big language models to intelligent security incident response and lays the foundation for the construction of an efficient and intelligent security defense system in the future.

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